CASE STUDY

CDAC ADDMISSION PROCESS

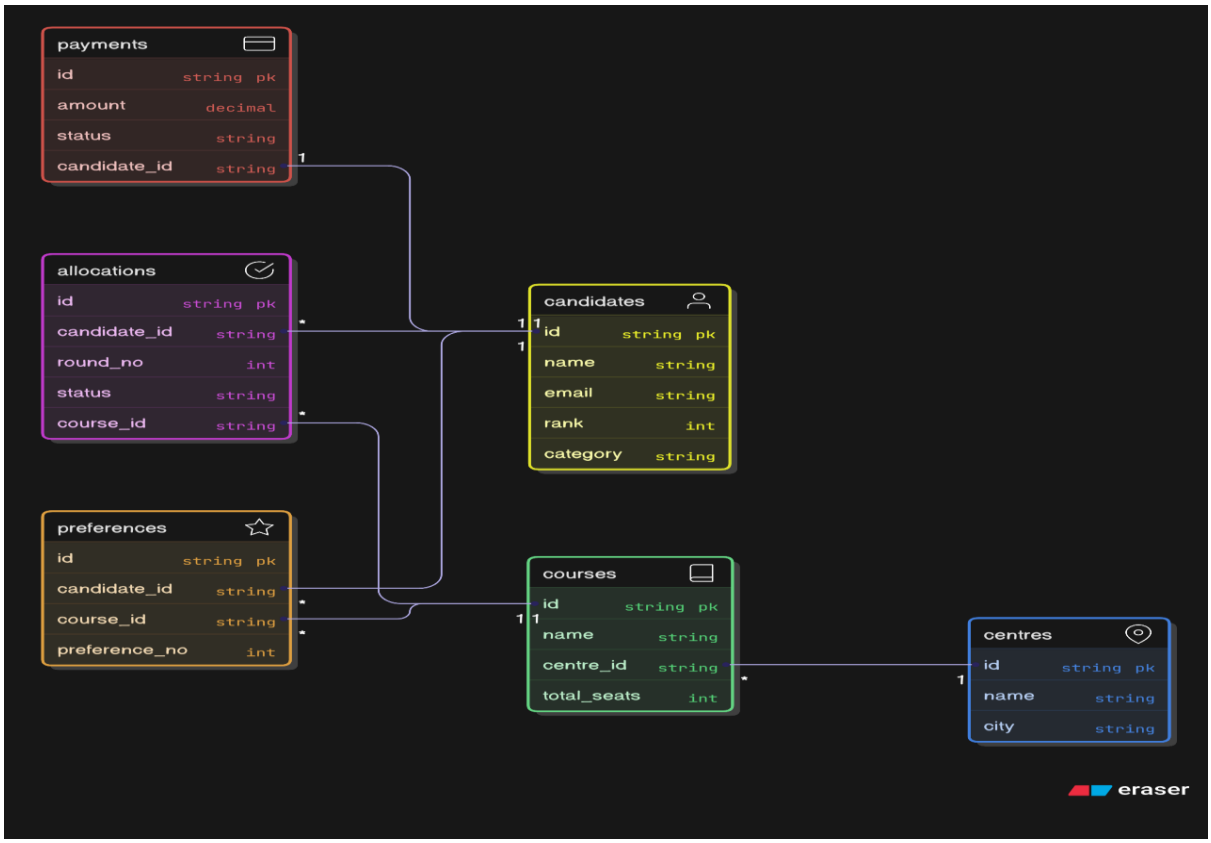
INTRODUCTION:

The CDAC Admission Portal is used to manage the admission process for various courses offered by CDAC. Students appear for the CCAT examination and receivea rank based on their performance. Based on the rank, students fill theircourse and center preferences. The system allocates seats according to rank,preferences, and seat availability. This case study designs a database system to manage the admission process using MySQL.

OBJECTIVES:

1. To design a database for CDAC admission system.
2. To store candidate information and CCAT ranks.
3. To manage course and center information.
4. To store candidate preferences.
5. To allocate seats based on rank and preference.

ER DIAGRAM



ENTITY DISCRIPTION:

Candidate

Stores candidate details.

Attributes

- Candidate_id (Primary Key)
- name

Centers

Stores CDAC center information.

Attributes

- centerid (Primary Key)
- center_name
- city

Courses

Stores course details.

Attributes

- courseid (Primary Key)
- course_name
- center_id
- total_seats

Preferences

Stores student course-center preferences.

Attributes

- preference_order
- candidate_id
- course_id
- pref_id

Allocation

Stores seat allocation results.

Attributes

- alloc_id
- candidate_id
- course_id
- round_no
- status

DATABASE TABLES (MySQL)

CANDIDATE TABLE

```
KD3_rutuja_96226>use AdmissionSystem;
Database changed
KD3_rutuja_96226>CREATE TABLE candidate (
    ->     candidate_id INT AUTO_INCREMENT,
    ->     name VARCHAR(100) NOT NULL,
    ->     email VARCHAR(100) UNIQUE NOT NULL,
    ->     c_rank INT NOT NULL CHECK (c_rank > 0),
    ->
    ->     PRIMARY KEY (candidate_id)
    -> );
Query OK, 0 rows affected (0.843 sec)

KD3_rutuja_96226>INSERT INTO candidate (name, email, c_rank)
    -> VALUES
    -> ('Rutuja', 'prerana@gmail.com', 1),
    -> ('Ankit', 'ankit@gmail.com', 2),
    -> ('Jasmine', 'jasmine@gmail.com', 3);
Query OK, 3 rows affected (0.375 sec)
Records: 3  Duplicates: 0  Warnings: 0

KD3_rutuja_96226>select * from candidate;
+--------------+---------+-------------------+--------+
| candidate_id | name    | email             | c_rank |
+--------------+---------+-------------------+--------+
|            1 | Rutuja  | prerana@gmail.com |      1 |
|            2 | Ankit   | ankit@gmail.com   |      2 |
|            3 | Jasmine | jasmine@gmail.com |      3 |
+--------------+---------+-------------------+--------+
3 rows in set (0.114 sec)
```

CENTER TABLE

```
KD3_rutuja_96226>CREATE TABLE center (
    -> center_id INT AUTO_INCREMENT PRIMARY KEY,
    -> center_name VARCHAR(100) NOT NULL,
    -> city VARCHAR(50));
Query OK, 0 rows affected (0.403 sec)

KD3_rutuja_96226>INSERT INTO center(center_name,city)
    -> VALUES
    -> ('CDAC Pune','Pune'),
    -> ('CDAC Mumbai','Mumbai'), ('CDAC Karad','Karad');
Query OK, 3 rows affected (0.088 sec)
Records: 3  Duplicates: 0  Warnings: 0

KD3_rutuja_96226>select * from center;
+-----------+-------------+--------+
| center_id | center_name | city   |
+-----------+-------------+--------+
|         1 | CDAC Pune   | Pune   |
|         2 | CDAC Mumbai | Mumbai |
|         3 | CDAC Karad  | Karad  |
+-----------+-------------+--------+
3 rows in set (0.011 sec)
```

COURSES TABLE

```
KD3_rutuja_96226>CREATE TABLE course (
    -> course_id INT AUTO_INCREMENT PRIMARY KEY,
    -> course_name VARCHAR(100),
    -> center_id INT,
    -> total_seats INT CHECK(total_seats > 0),
    -> FOREIGN KEY (center_id)
    -> REFERENCES center(center_id));
Query OK, 0 rows affected (0.487 sec)
```

```
KD3_rutuja_96226>INSERT INTO course(course_name,center_id,total_seats)

    -> VALUES
    ->  ('DAC',1,10),
    ->  ('DBDA',1,1),
    ->  ('DAC',2,1),
    ->  ('AI',3, 10);
Query OK, 4 rows affected (0.122 sec)
Records: 4  Duplicates: 0  Warnings: 0

KD3_rutuja_96226>select * from course;
+----------+-------------+----------+-------------+
| course_id | course_name | center_id | total_seats |
+----------+-------------+----------+-------------+
|        11 | DAC         |        1 |          10 |
|        12 | DBDA        |        1 |           1 |
|        13 | DAC         |        2 |           1 |
|        14 | AI          |        3 |          10 |
+----------+-------------+----------+-------------+
4 rows in set (0.013 sec)
```

PREFERANCE TABLE

```
KD3_rutuja_96226>CREATE TABLE preference (
    -> pref_id INT AUTO_INCREMENT PRIMARY KEY,
    -> candidate_id INT,
    -> course_id INT,
    -> preference_order INT CHECK(preference_order BETWEEN 1 AND 5),
    -> UNIQUE(candidate_id, preference_order),
                FOREIGN KEY(candidate_id) REFERENCES candidate(candidate_id),
    -> FOREIGN KEY(course_id) REFERENCES course(course_id));
Query OK, 0 rows affected (0.859 sec)
```

```
KD3_rutuja_96226>INSERT INTO preference (candidate_id, course_id, preference_order)
    -> VALUES
    -> (1,11,1),
    -> (1,12,2),
    -> (2,11,1),
    -> (3,13,1);
Query OK, 4 rows affected (0.165 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
KD3_rutuja_96226>select * from preference;
+---------+--------------+-----------+------------------+
| pref_id | candidate_id | course_id | preference_order |
+---------+--------------+-----------+------------------+
|      11 |            1 |        11 |                1 |
|      12 |            1 |        12 |                2 |
|      13 |            2 |        11 |                1 |
|      14 |            3 |        13 |                1 |
+---------+--------------+-----------+------------------+
4 rows in set (0.015 sec)
```

ALLOCATION TABLE

```
KD3_rutuja_96226>select * from allocation;
+----------+--------------+-----------+----------+-----------+
| alloc_id | candidate_id | course_id | round_no | status    |
+----------+--------------+-----------+----------+-----------+
|        1 |            1 |        11 |        1 | ALLOCATED |
|        2 |            2 |        11 |        1 | ALLOCATED |
|        3 |            3 |        13 |        1 | ALLOCATED |
|        4 |            1 |        11 |        1 | ALLOCATED |
|        5 |            2 |        11 |        1 | ALLOCATED |
|        6 |            1 |        11 |        1 | ALLOCATED |
|        7 |            2 |        11 |        1 | ALLOCATED |
|        8 |            3 |        13 |        1 | ALLOCATED |
+----------+--------------+-----------+----------+-----------+
8 rows in set (0.010 sec)
```

STORED PROCEDURE (Round 1 Allocation)

This procedure allocates seats **based on rank order**

Seat Allocation Logic

The system allocates seats using the following logic:

1. Candidates are sorted based on CCAT rank.
2. The system checks the candidate's first preference.
3. If a seat is available, the seat is allocated.
4. If the seat is not available, the system checks the next preference.
5. The allocated seat is stored in the Allocation table.

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```
KD3_rutuja_96226>DELIMITER //
KD3_rutuja_96226>
KD3_rutuja_96226>CREATE PROCEDURE round1_allocation1()
    -> BEGIN
    ->
    -> DECLARE done INT DEFAULT 0;
    -> DECLARE c_id INT;
    -> DECLARE p_course INT;
    ->
    -> DECLARE cur CURSOR FOR
    -> SELECT candidate_id, course_id
    -> FROM preference
    -> WHERE preference_order = 1
    -> ORDER BY candidate_id;
    ->
    -> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    -> OPEN cur;
    ->
    -> read_loop: LOOP
    ->
    -> FETCH cur INTO c_id, p_course;
    ->
    -> IF done THEN
    -> LEAVE read_loop;
    -> END IF;
    ->
    -> IF (SELECT total_seats FROM course WHERE course_id = p_course) > 0 THEN
    ->
    -> INSERT INTO allocation(candidate_id,course_id,round_no)
    -> VALUES(c_id,p_course,1);
    ->
    -> UPDATE course
    -> SET total_seats = total_seats - 1
    -> WHERE course_id = p_course AND total_seats > 0;
    ->
    -> END IF;
    ->
    -> END LOOP;
    ->
    -> CLOSE cur;
    ->
    -> END //
Query OK, 0 rows affected (0.132 sec)
```

FINAL OUTPUT

```
KD3_rutuja_96226>select * from allocation;
+----------+--------------+-----------+----------+-----------+
| alloc_id | candidate_id | course_id | round_no | status    |
+----------+--------------+-----------+----------+-----------+
|        1 |            1 |        11 |        1 | ALLOCATED |
|        2 |            2 |        11 |        1 | ALLOCATED |
|        3 |            3 |        13 |        1 | ALLOCATED |
|        4 |            1 |        11 |        1 | ALLOCATED |
|        5 |            2 |        11 |        1 | ALLOCATED |
|        6 |            1 |        11 |        1 | ALLOCATED |
|        7 |            2 |        11 |        1 | ALLOCATED |
|        8 |            3 |        13 |        1 | ALLOCATED |
+----------+--------------+-----------+----------+-----------+
8 rows in set (0.010 sec)
```

CONCLUSION

The CDAC Admission Portal database system efficiently manages the entire admission process. It stores candidate information, exam results, course and center details, preferences, and seat allocation results. The system ensures fair seat allocation based on rank and candidate preferences.